Saint Kentigern College	**Level 1 Exploring Data 2024**			**N**	Clear fail	
AS Number	91944 v3	**Credits**	5			
Name				**A**	All required	
Marker				**M**	All required	
Check Marker				**E**	All required. (Excellence is exceptional)	

Comparison Investigation

	Achieved **(all required)**		**Merit** **(all required)**		**Excellence** **(all required)**	
Problem & Plan	Actively participated in planning & collecting data		Gives a purpose (who cares and why), or a reasoned hypothesis.		**TWO instances of contextual insight** This could include: ❑ A fully reasoned hypothesis (explains why in context) ❑ Possible biases ❑ Other insightful reflections	
	Explained two different sources of variation in the data collection process: ❑ Natural or real ❑ Occasion-to-occasion ❑ Measurement ❑ Induced ❑ Sampling		Investigation question must include: ❑ Numeric variable ❑ Categorical variable ❑ "tends" ❑ Direction ❑ Population (Either the **teacher or student** can form a question)			
Data	Display: (see note on bottom) ❑ Sample 100 per group ❑ Box plot		Displays ❑ Dot and box plot ❑ Summary Statistics		**TWO instances of statistical insight** This could include: ❑ Effect of not managing sources of variation on overall findings ❑ Extending analysis beyond the obvious features ❑ Other insightful reflections	
Analysis	**Two** features **described** in **context** (refers to **both** numeric variables in each feature): ❑ Shape ❑ Centre ❑ Spread ❑ Shift and overlap ❑ Unusual features		**Two** features **visually justified** in **context**: ❑ Shape ❑ Centre ❑ Spread ❑ Shift and overlap ❑ Unusual features			
Conclusion & Report			Answers the investigation question (based on the DBM – OVS rule). ❑ Numeric variable ❑ Categorical variable ❑ "Tends" ❑ Direction ❑ Population.		At all times the report is: ❑ Concise and logical ❑ Clearly communicated in context through entire investigation ❑ Uses contextually correct statistical terminology ❑ Correctly referencing graphs and measures (including units)	
			Conclusion **justified** with reference to the appropriate rule (eg ¾-½ or DBM – OVS)			

Note: Students can take a sample of between 20-40 per group, but must then use the ½ - ¾ rule in their conclusion, followed by the DBM/OVS if insufficient evidence.

Relationship Investigation

	Achieved (all required)		Merit (all required)		Excellence (all required)	
Problem & Plan	Actively participated in planning & collecting data		Gives a purpose (who cares and why), or a reasoned hypothesis.		**TWO instances of contextual insight** This could include: ❑ A fully reasoned hypothesis (explains why in context) ❑ Possible biases ❑ Other insightful reflections	
	Explained two different sources of variation in the data collection process: ❑ Natural or real ❑ Occasion-to-occasion ❑ Measurement ❑ Induced ❑ Sampling		Investigation question must include: ❑ Explanatory variable ❑ Response variable ❑ Participants (Either the **teacher or student** can form a question)			
Data	Display: ❑ Minimum 30 pairs of data ❑ Scatter graph		Displays: ❑ Visually fitted line of best fit		**TWO instances of statistical insight** This could include: ❑ Effect of not managing sources of variation on overall findings ❑ Extending analysis beyond the obvious features ❑ Other insightful reflections	
Analysis	**Two** features **described** in **context** (refers to **both** numeric variables in each feature): ❑ Trend ❑ Direction ❑ Strength ❑ Variation ❑ Unusual features (clusters/outliers etc)		**Two** features **visually justified** in **context**: ❑ Trend ❑ Direction ❑ Strength ❑ Variation ❑ Unusual features (clusters/outliers etc)			
Conclusion & Report			Visual prediction described: ❑ Explanatory variable (including units) ❑ Response variable (including units ❑ Participants		At all times the report is: ❑ Concise ❑ Clearly communicated in context through entire investigation ❑ Uses contextually correct statistical terminology ❑ Correctly referencing graphs and measures (including units)	

Time Series Investigation

	Achieved (all required)		Merit (all required)		Excellence (all required)	
Problem & Plan	Actively participated in planning & collecting data		Gives a purpose (who cares and why), or a reasoned hypothesis.		**TWO instances of contextual insight** This could include: ❑ A fully reasoned hypothesis (explains why in context) ❑ Possible biases ❑ Other insightful reflections	
	Explained two different sources of variation in the data collection process: ❑ Natural or real ❑ Occasion-to-occasion ❑ Measurement ❑ Induced ❑ Sampling		Investigation question must include: ❑ Time variable (including units) ❑ Numeric variable (including units) ❑ Participants (Either the **teacher or student** can form a question)			
Data	Display: ❑ Minimum 5 cycles ❑ Time series graph		Display: ❑ Long-term trend line added ❑ Visually fitted predictions		**TWO instances of statistical insight** This could include: ❑ Effect of not managing sources of variation on overall findings ❑ Extending analysis beyond the obvious features ❑ Other insightful reflections	
Analysis	**Two** features **compared** in **context** (refers to **both** variables in each feature): ❑ Trend ❑ Seasonality / Cycle ❑ Patterns		**Two** features **visually justified** in **context**: ❑ Trend ❑ Seasonality / cycle ❑ Patterns			
Conclusion & Report			Visual prediction described: ❑ Time variable (including units) ❑ Numeric variable (including units) ❑ Participants		At all times the report is: ❑ Concise ❑ Clearly communicated in context through entire investigation ❑ Uses contextually correct statistical terminology ❑ Correctly referencing graphs and measures (including units)	

Experimental Probability Investigation

	Achieved (all required)		Merit (all required)		Excellence (all required)	
Problem & Plan	Actively participated in planning & collecting data		Gives a purpose (who cares and why), or a reasoned hypothesis.		**TWO instances of contextual insight** This could include: ❑ A fully reasoned hypothesis (explains why in context) ❑ Possible biases ❑ Other insightful reflections	
Problem & Plan	Explained two different sources of variation in the data collection process: ❑ Natural or real ❑ Occasion-to-occasion ❑ Measurement ❑ Induced ❑ Sampling		Investigation question must include: ❑ Independent variable ❑ Response variable ❑ Participants (Either the **teacher or student** can form a question)			
Data	Display: ❑ 30 trials or more ❑ Table or bar graph		Displays: ❑ Two-way table ❑ Bar graph		**TWO instances of statistical insight** This could include: ❑ Effect of not managing sources of variation on overall findings ❑ Extending analysis beyond the obvious features ❑ Other insightful reflections	
Analysis	**Two** features **compared** in **context** (refers to **both** numeric variables in each feature): ❑ Clusters ❑ Centre ❑ Spread ❑ Shape		**Two** features **visually justified** in **context**: ❑ Clusters ❑ Centre ❑ Spread ❑ Shape			
Conclusion & Report			Answers the investigation question: ❑ Independent variable ❑ Response variable ❑ Participants		At all times the report is: ❑ Concise ❑ Clearly communicated in context through entire investigation ❑ Uses contextually correct statistical terminology ❑ Correctly referencing graphs and measures (including units)	